Extra Exercises 01 – What Is AI?

Goes with cheatsheet `01_what_is_ai`. All tasks work with any AI service, including a free plan.

Safety rule: Use only invented or anonymized data. Suggested solutions are at the end. Only look once you're done.

Task 1 ★ Getting started: Match the terms

Scenario: A colleague says: "It's all the same thing anyway: AI, machine learning, LLM."

Task: Match each term to the fitting description.

Term	Description
1. AI	A. The program learns from many examples instead of being given fixed rules.
2. Machine learning	B. A language model that produces text by predicting the most likely next word.
3. LLM	C. Umbrella term for programs that solve tasks that would otherwise require human thinking.

Hint: Go from general to specific. An LLM is a special case.

Task 2 ★★ Everyday: What is the AI good for?

Scenario: You have six tasks on your list.

Task: Sort each task into a group: - **Well suited:** AI can do most of the work - **Only with checking:** AI helps, but you verify the result - **Better solved differently:** Use a different tool or source

Tasks: 1. Write a birthday greeting for Person A 2. Summarize a long newsletter in five sentences 3. Know whether the bakery around the corner is open today 4. Calculate an invoice with several items exactly 5. Brainstorm three ideas for a weekend with friends 6. Name the source for a statistic in a presentation

Hint: Think of the strengths and weaknesses from the cheatsheet. Ask yourself: What happens if the answer is wrong?

Task 3 ★★ Everyday: Confident, but wrong?

Scenario: You want to find out how the AI handles something that doesn't exist.

Task: Ask your AI this question. The book is entirely made up.

```
Summarize the novel "The Moon Garden of Person A" for me in three
↪  sentences.
```

Assess the answer: - Does the AI admit it doesn't know the book? - Or does it invent a plot?

Then ask a follow-up question:

```
How do you know that? How sure are you?
```

Hint: Both outcomes are instructive. If the AI invents something, you see how convincing nonsense can sound. If it answers honestly, you see what a good result looks like.

Task 4 ★★★ Challenge: Chatbot, search engine, human

Scenario: You're planning an outing with your family on the weekend.

Task: Ask the same question in three places: 1. a search engine 2. an AI chatbot 3. a human (friend, colleague, neighbor)

```
Which three excursion destinations are suitable for a weekend with
↪  children in [your region]?
```

Fill in the table in your own words:

Criterion	Search engine	Chatbot	Human
How up to date is the answer?			
Can I verify the source?			
How well does it fit me?			
Who is responsible?			

Finally, write one sentence: For which question will you use which tool in the future?

Hint: There is no single "best" source. Some chatbots also use a web search. Note whether yours does.

Suggested solutions

Task 1: 1 → C, 2 → A, 3 → B. AI is the umbrella term, machine learning is one way of getting there, and an LLM is a form of it for language.

Task 2: Possible classification: - *Well suited:* 1 (birthday greeting), 2 (summary; with a short text it's easy to check against the original), 5 (brainstorming ideas) - *Only with checking:* 2 for important content, because details can be missing or wrong - *Better solved differently:* 3 (current opening hours: website or phone call), 4 (exact invoice: calculator or spreadsheet), 6 (source: find the original source; the AI can invent sources)

What matters is your reasoning: The more consequential a mistake, the more you check.

Task 3: Two outcomes are possible. - *The AI admits it doesn't know the book:* That's the desired behavior. You can deliberately encourage such answers, e.g. with "Say 'I don't know' when you're unsure." - *The AI invents a plot:* That's a hallucination — an invented but convincingly sounding answer. Remember: A confident tone says nothing about correctness. More on this in cheatsheet 08.

Even a follow-up question about confidence doesn't guarantee truth. Verify in an independent source.

Task 4: Typical observations (yours may differ): - *Search engine:* usually current, sources visible, you have to select and read yourself. - *Chatbot:* fast and well formulated, but currency and sources depend on whether a web search is used. Verify claims. - *Human:* knows you and your region, but has only a limited overview. Responsibility lies with you; a human can ask follow-up questions and contribute experience.

A sensible closing sentence could be: "For current facts I use the search engine, for ideas and wording the chatbot, and for personal decisions I additionally ask a human."

Next: Cheatsheet 02 (Tokens & Context Window), Test A (`03_tests/test_A_foundation`)

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